THE PROJECTION FORMULA AND COMPOSITION OF PUSHFORWARDS FOR A CLASSICAL WEIL COHOMOLOGY THEORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the projection formula obtained by pairing with an arbitrary class and the composition of pushforwards reduced to the functoriality of pullback.

Editorial note. The label lemma-pushforward-classical names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-pushforward-classical –start 1603 –end 1624 –local-id 28170 returned RESULT=OK. Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The pairing given by the degree map is perfect. Axiom (A) describing the nondegenerate pairing between complementary cohomology groups through the degree map. Allows replacing the equality of degree pairings by the equality of the classes involved.
- Pullback is compatible with the degree map. Axiom (A)(b) together with the compatibility of pullback with the pairing, so that the degree of a pullback is the degree of the original class. Moves pullbacks across the degree sums in the two displayed computations.

Lemma 0.1. *Assume given (D1) and (D3) satisfying (A). For $f : X \to Y$ a morphism of smooth projective varieties we have $f_*(f^*b \cup a) = b \cup f_*a$. If $g : Y \to Z$ is a second morphism of smooth projective varieties, then $g_* \circ f_* = (g \circ f)_*$.*

Proof. The first equality holds because

$$\int_Y c \cup b \cup f_*a = \int_X f^*c \cup f^*b \cup a = \int_Y c \cup f_*(f^*b \cup a).$$

The second equality holds because

$$\int_Z c \cup (g \circ f)_*a = \int_X (g \circ f)^*c \cup a = \int_X f^*g^*c \cup a = \int_Y g^*c \cup f_*a = \int_Z c \cup g_*f_*a$$

This ends the proof. □

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